Exercise Sheet (2): Introduction to Pneumatics & Hydraulics Control

Course: Fluid Power Systems (MEC 431) - BUC
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Question 1 & 3: Simple Hydraulic Circuit & Elements Function

> **TA Note:** Questions 1 and 3 are conceptually identical. Q3 asks for a specific list of elements (pump, DCV, cylinder, relief valve, gauge) and power transformation discussion.

Circuit Diagram Description:

1. **Reservoir (Tank):** Stores the hydraulic fluid and cools it.
2. **Filter:** Cleans the fluid from solid contaminants.
3. **Pump:** Converts mechanical power (from electric motor) to hydraulic power (flow under pressure).
4. **Pressure Relief Valve (PRV):** Protects the system from overpressure by diverting flow back to the tank when the pressure setting is reached.
5. **Directional Control Valve (DCV):** Controls the direction of fluid flow (e.g., 4/3-way valve) to extend, retract, or stop the cylinder.
6. **Actuator (Hydraulic Cylinder):** Converts hydraulic power back into mechanical power (linear motion and force).

Power Transmission and Transformation:

- **Input:** Electric Motor (Electrical Energy → Mechanical Energy).
- **Transformation 1:** Hydraulic Pump (Mechanical Energy → Hydraulic Energy).
- **Transmission:** Pipes & Valves carry the pressurized fluid.
- **Transformation 2:** Hydraulic Cylinder (Hydraulic Energy → Mechanical Energy).

Question 2: Advantages and Disadvantages of Hydraulic

Systems

Advantages:

1. High power-to-weight ratio (High power density).
2. Smooth, continuous operation and easy reversal of direction.
3. Precise control of force and speed (stepless variation).
4. Self-lubricating system (oil reduces wear).
5. Safe in hazardous environments (no sparks compared to electrical).

Disadvantages:

1. Risk of oil leakage causing mess or hazards.
2. Fire hazard if standard mineral oils are used near ignition sources.
3. Requires precise filtration (sensitive to dirt and contamination).
4. Lower overall efficiency compared to pure mechanical transmission (due to fluid friction and throttling losses).

Question 4: Cylinder Extension Mode (Standard)

Given Data:

- Delivery line pressure $P = 195 \text{ bar} = 195 \times 10^5 \text{ Pa}$
- Pump flow rate $Q_P = 39 \text{ L/min} = \frac{39 \times 10^{-3}}{60} \text{ m}^3/\text{s} = 6.5 \times 10^{-4} \text{ m}^3/\text{s}$
- Piston diameter $D = 90 \text{ mm} = 0.09 \text{ m}$
- Piston rod diameter $d = 50 \text{ mm} = 0.05 \text{ m}$

1. Calculate Areas:

- **Piston Area (A_p):**
  $A_p = \frac{\pi}{4} D^2 = \frac{\pi}{4} (0.09)^2 = 6.3617 \times 10^{-3} \text{ m}^2 = 63.62 \text{ cm}^2$
- **Rod Area (A_r):**
  $A_r = \frac{\pi}{4} d^2 = \frac{\pi}{4} (0.05)^2 = 1.9635 \times 10^{-3} \text{ m}^2 = 19.63 \text{ cm}^2$
- **Annulus Area (A_{an}):**
  $A_{an} = A_p - A_r = (6.3617 - 1.9635) \times 10^{-3} = 4.3982 \times 10^{-3} \text{ m}^2 = 43.98 \text{ cm}^2$

2. Loading Force (F):

(Assuming $P_T \approx 0$ on the annulus side)

$F = P \times A_p = (195 \times 10^5) \times (6.3617 \times 10^{-3}) = 124,053 \text{ N}$

F = 124.05 kN

3. Piston Speed (v):

$v = \frac{Q_P}{A_p} = \frac{6.5 \times 10^{-4}}{6.3617 \times 10^{-3}} = 0.1022 \text{ m/s}$

v = 10.22 cm/s

4. Returned Flow Rate (Q_T):

$Q_T = v \times A_{an} = 0.1022 \times (4.3982 \times 10^{-3}) = 4.495 \times 10^{-4} \text{ m}^3/\text{s}$

$$Q_T = 4.495 \times 10^{-4} \times 1000 \times 60$$

Q_T = 26.97 L/min

5. Pump Output Hydraulic Power (N_h):

$$N_h = P \times Q_P = (195 \times 10^5) \times (6.5 \times 10^{-4}) = 12,675 \text{ W}$$

N_h = 12.675 kW

6. Cylinder Output Mechanical Power (N_m):

$$N_m = F \times v = 124,053 \times 0.1022 = 12,675 \text{ W}$$

N_m = 12.675 kW

> **Note:** $N_m = N_h$ because the problem implies neglecting losses (Efficiency = 100%).

Question 5: Cylinder Extension Mode (Differential Connection)

> **TA Note for Students:** Clarify the typo in the sheet; "problem 6" actually refers to "problem 4". In a differential connection, the rod end oil is rerouted to the blank end instead of the tank. This means the pump only has to fill the volume of the **ROD (A_r)**, not the whole piston area.

1. Piston Speed (v):

In differential mode, effective area = A_r.

$v = \frac{Q_P}{A_r} = \frac{6.5 \times 10^{-4}}{1.9635 \times 10^{-3}} = 0.331 \text{ m/s}$

v = 33.1 cm/s

2. Inlet Flow Rate (Q_{in}):

$Q_{in} = v \times A_p = 0.331 \times 6.3617 \times 10^{-3} = 2.106 \times 10^{-3} \text{ m}^3/\text{s}$

Q_{in} = 126.36 L/min

(Check: $Q_{in} = Q_P + Q_{out} = 39 + 87.36 = 126.36$)

3. Returned Flow Rate (Q_{out}):

$Q_{out} = v \times A_{an} = 0.331 \times 4.3982 \times 10^{-3} = 1.456 \times 10^{-3} \text{ m}^3/\text{s}$

Q_{out} = 87.36 L/min

4. Loading Force (F):

Pressure acts on BOTH sides of the piston!

$F = P \times A_p - P \times A_{an} = P \times (A_p - A_{an}) = P \times A_r$

$F = (195 \times 10^5) \times (1.9635 \times 10^{-3}) = 38,288 \text{ N}$

F = 38.29 kN

5. Pump Power (N_h) & Cylinder Power (N_m):

- $N_h = 12.675\ \text{kW}$ (Same as Q4, pump operates at same P and Q)
- $N_m = F \times v = 38,288 \times 0.331 = 12,673\ \text{W}$ → **N_m ≈ 12.67 kW**

Comments on Calculation Results (Comparison Q4 vs Q5):

- **Speed:** Increased by a ratio of $\frac{A_p}{A_r} = \frac{63.62}{19.63} \approx 3.24$ times.
- **Force:** Decreased by the exact same ratio ($\frac{1}{3.24}$).
- **Power:** Remains constant. The differential connection acts like a mechanical gear ratio: it trades force for speed while keeping the transmitted power constant.